

Ansh Srivastava

Software Developer · Backend Systems
· ML Enthusiast

Contact

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Bangalore, India

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Education

BMS College of Engineering

BE — Information Science Engg.
2023–2027 · CGPA: 9.25

Kolhapur Public School

XII — CBSE · 94%
2021–2023

Shantiniketan School

X — CBSE · 92%
2019–2021

Skills

Programming Languages

C++ · Java · Python · C

Backend

MERN

Core CS

DSA · OS · DBMS · Computer Networks

Databases

MySQL · MongoDB

Tools

Git · Linux · Jupyter · Overleaf

Arduino IDE

ML / AI

PyTorch · TensorFlow · Keras

Languages

English Fluent

Hindi Native

Interests

Writing Poetry

Music

Novels & Manga

Exploring new cities

BE · Information Science Engineering

B.M.S College of Engineering, Bangalore

2023 – 2027 · CGPA: 9.25

Professional Experience

Samsung PRISM R&D Intern, Samsung Research Institute

Jan 2026 – Present | Bangalore, India

- Working on **distributed multimedia systems** for real-time communication.
- Contributing to the design and development of **distributed systems** for real-time audio/video processing across multiple devices.
- Gaining hands-on experience in **networking protocols, Android system-level development, and performance optimization**.

Research

ThoraxSense: Multi-Disease Detection on Chest X-Rays, Nature's Scientific Reports

Submitted 2025 | Under Review

- Designed a resource-efficient multi-label classification framework for 14 thoracic diseases on 112,120 chest X-rays, achieving **micro AUROC 0.8406** and **mean AUROC 0.8003**.
- Implemented deep learning pipelines using **DenseNet121 (PyTorch)** and **VGG16 (TF/Keras)** with optimization techniques including gradient clipping, weighted BCE loss, and adaptive learning rate scheduling.
- Evaluated cross-dataset generalisation on CheXpert and applied **Grad-CAM** for model interpretability.

Projects

Student Academic ERP System

· Full Stack Web Application

Streamlit · SQLite · Python

- Developed a **Student ERP web application** for managing academic records, attendance, and analytics.
- Designed backend workflows for **data ingestion and validation** with bulk CSV upload support.
- Built modular components and dashboards for real-time insights and reporting.

Order Karo — Full-Stack Food Ordering Platform

· MERN Stack

MongoDB · JWT Auth

- Developed a full-stack food ordering platform using the MERN stack with secure **JWT-based authentication** and user session management.
- Designed and implemented **RESTful APIs** for order processing, menu management, and user workflows with a modular backend architecture.
- Structured database schemas and relationships for efficient data handling, ensuring maintainability and scalability of the system.

AI & IoT Framework for Carbon Credit Farming

[In Progress] · IoT · ML ·

Python IoT Sensors (DHT11, Soil Moisture, pH) · ESP32 · Random Forest · XGBoost · Sentinel-2 · Cloud Dashboard

- Developing a scalable IoT-cloud-ML pipeline to estimate **Soil Organic Carbon (SOC)** for carbon credit farming; integrating ESP32 soil sensors with Sentinel-2 satellite imagery and ML models (XGBoost, Random Forest) for farm-specific predictions.
- Designing a **Soil Health / Carbon Sequestration Potential Index** and interactive dashboard for real-time visualisation of sensor data, environmental indicators, and model predictions; targeting automated, low-cost SOC monitoring at farm scale.

Relevant Coursework

Data Structures & Algorithms · Operating Systems (Process Scheduling, Memory Management) · Database Management Systems (SQL, NoSQL) · Computer Networks · Object-Oriented Programming (C++/Java) · Software Engineering · Theory of Computation · Machine Learning · Linear Algebra & Probability